

Quantum Support Vector Machine-Based Approach for Fruit Ripeness Detection

Prantik Sarkar¹, Qadir Eqbal², Parth Pandey³, Subhankar Joardar⁴

^{1,2,3} Student, CSE, Haldia Institute Of Technology
prantiksarkar400@gmail.com, qadiriqbal74@gmail.com, parthpandey201@gmail.com,
⁴ Professor, Haldia Institute Of Technology
subhankarranchi@yahoo.co.in

Abstract. The advancement of quantum computing offers potential advantages in parallelism and kernel-based methods. Our project applies a Quantum Support Vector Machine (QSVM) for the task of classifying the ripeness of blackberries using computer vision techniques. The traditional fruit ripeness detection methods in agriculture often rely on manual inspection otherwise classical machine learning models. This model requires large datasets and extensive preprocessing. This project proposes a hybrid classical-quantum approach that integrates classical feature extraction from images with quantum-enhanced classification.

Images of blackberries are categorized into two classes such as unripe and ripe. Then they are processed using OpenCV to extract HSV-based colour histograms. These histograms capture the color profile that changes significantly as blackberries ripen. To keep the feature space small and suitable for current quantum processors, only the first two normalized histogram values were used. In our QSVM model we use Qiskit's QuantumKernel with a ZZFeatureMap to transform classical image features into quantum states. This allows the model to measure similarities between samples in a quantum-enhanced way. We trained and tested the model using Qiskit's qasm_simulator with each quantum circuit of 1024 times to ensure stable results. We evaluated our model with classification reports and confusion matrix.

Keywords: Quantum Support Vector Machine, QSVM, Quantum Machine Learning, Fruit Ripeness Detection, Blackberry Classification, Computer Vision, Image Processing, HSV Colour Histogram, Qiskit, Quantum Kernel, ZZFeatureMap, QASM Simulator, Hybrid Quantum-Classical Model, Agricultural Technology, Feature Extraction, Quantum Computing in Agriculture.

1 Introduction

Fruits play a crucial role in our life. They are an essential part of a healthy and balanced diet due to their vitamins, minerals and fiber. They play a vital role in preventing chronic diseases and support overall well-being. Various fruits contain vitamins like A, C and minerals like potassium, iron etc. Various antioxidants in fruits help to nourish skin and hair. To get these nutrients at proper quantity it is also essential to consume those fruits at proper ripeness state. Various models and ML, DL technologies are there to detect the fruit ripeness state, i.e. ML methods such as SVM, K-Nearest Neighbour algorithm (KNN), Decision Tree – Some Deep Learning (DL) technologies such as Artificial Neural Networks (ANNs), Convolution Neural Networks (CNNs), and also Quantum algorithms like Quantum Support Vector Machine (QSVM) using quantum kernel etc. provide very pleasing results.

Classifying fruits using a specific model poses significant challenges due to the wide variations in shape, size, colour and texture across different fruit types. For instance, although Fuji apples and Granny Smith apples share a similar shape, they differ considerably in colour and texture. This diversity makes it difficult to design a one-size-fits-all classification model. As a result, both the upstream task of fruit recognition and the downstream task of ripeness classification present considerable complexity. On the other hand, it is essential to enable automation within the supply chain of fruit crops to keep up with the worldwide demand for high quality, safe food [1].

We have implemented a Quantum Support Vector Machine (QSVM) in our study instead of traditional methods like CNN or AlexNet. Though the traditional methods provide great results but they need complex computational resources and large no. of datasets which makes it a bit hard to implement. On the other hand, QSVM offers a more efficient and scalable solution.

Quantum bits (qubits), which can exist in multiple states simultaneously through superposition and influence one another via entanglement, are the building blocks of quantum algorithms. Because of these special characteristics, Quantum Machine Learning (QML) algorithms can complete some calculations more quickly and effectively than conventional machine learning models.

QSVM serves as the quantum counterpart of the classical Support Vector Machine (SVM), a supervised learning algorithm that finds an optimal hyperplane to separate data points from different classes. QSVM leverages quantum kernel estimation to compute the similarity between data points within a high-dimensional quantum feature space, enhancing classification performance.

1.1 Key Features of QSVM

- It uses quantum kernel methods that map classical data into a quantum Hilbert space.

- It uses quantum feature maps (e.g. ZZFeatureMap) to transform low-dimensional data into highly expressive quantum states.
- It leverages quantum circuits to capture complex relationships between input features.
- It can be simulated using Qiskit's qasm_simulator which enable a scope of experimentation without real quantum hardware.

1.2 QSVM Over CNN or AlexNet

Table 1. CNN/AlexNet Vs QSVM

Criteria	CNN / AlexNet	QSVM
Data Requirements	Requires large labeled datasets	Works on small datasets
Computational Resources	High (needs GPUs/TPUs)	Lower for small datasets (even on CPU via simulators)
Feature Extraction	Automatically extracts spatial features	Needs pre-extracted features (we use HSV histograms)
Explainability	Difficult to interpret deep layers	Easier to interpret in kernel space
Use Case	Best for complex, high-resolution image data	Suitable for small, low-dimensional problems

In this paper QSVM is used because the dataset is small and can manually be organized (ripe vs unripe. The input feature size is in low-dimensional (just 2 values per image). Training CNNs or AlexNet would require more computation time and GPU resources, whereas QSVM is lightweight and experimentally quantum-ready. Thus QSVM is a more efficient and educational choice for this specific task.

1.3 Features Used in This Paper

The classification task is based on colour features extracted from each fruit image. These features are:

1. HSV Histogram Features : Images are converted from BGR to HSV colour space. A 3D colour histogram (with 4 bins per channel) is computed. This results in a 64-length feature vector. We kept only the first 2 histogram values to match the 2D input required by the quantum feature map. Example's = cv2.calcHist([hsv], [0, 1, 2], None, [4, 4, 4], [0, 180, 0, 256, 0, 256]) features = hist[:2]
2. Standardization: All extracted features are scaled using StandardScaler(), so that they have a mean of 0 and unit variance. This helps QSVM make better predictions since the kernel is sensitive to input scale.
3. Binary Labels: Ripe fruits are labelled as 1. Unripe fruits are labelled as 0. This makes it a binary classification problem

4. Train-Test Split: 70% of the data is used for training, 30% is used for testing
5. Quantum Kernel Setup: ZZFeatureMap(feature_dimension=2, reps=2) maps classical features into a quantum space. The kernel is calculated on Qiskit's qasm_simulator.

1.4 Significance

Our QSVM model provides a real-world use of quantum machine learning. This model shows that how quantum machine learning techniques can handle a small scale and low feature dataset rather than relying on complex deep learning models. Few ML techniques like QSVM may provide scalable benefits in fields that demand quick, precise, and resource-efficient models like quantum hardware advances.

1.5 Accuracy Achieved

We found that the QSVM model had a test accuracy of roughly 85% after training and testing it on the dataset. This shows that 85% of the fruit photos were correctly classified as either ripe or unripe by the QSVM model. This accuracy shows how well quantum kernel-based learning works for small-scale classification tasks, even though the features used are simple (only two per image).

2 Related work

Over the years, a number of researchers have put forth various techniques to automate the process of determining the ripeness of fruit. Nevertheless, the majority of these methods rely on conventional machine learning models. Two CNN models were proposed by the authors in [1] to evaluate the quality of bananas. One model detected banana rot with 98.3% accuracy and general classification with 93.4% accuracy. Likewise, a five-layer CNN banana served as the basis for the model in [2]. These CNN models demonstrated the efficacy of hybrid approaches by achieving classification accuracy of up to 96.98% when paired with conventional classifiers such as K-Nearest Neighbor (KNN) and Random Forest (RF). In [3] a MobileNetV2 based deep learning model achieved training and validation accuracy of more than 99%. This suggests that a lightweight CNN architecture can also perform fruit classification. In the same way the authors of [4] trained a deep learning model on apple images and achieved accuracy of 99% for training and 98.8% for validation. For sorting and grading of mangoes [5] a CNN based approach uses few image augmentation techniques. They achieved 99.2% sorting, 96.7% grading accuracy among VGG16, ResNet152 and Inception V3. Transfer learning was used to assess existing models, including ResNet18, Inception-v3, and AlexNet for the stages of mulberry ripeness [6]. Among various ripeness stages and genotypes the ResNet18 model achieved the accuracy of 98.03%. Extended AlexNet (E-AlexNet) and similar deep learning are used for strawberry quality [7] detection which achieved the accuracy of 95.75%. The transfer learning technique for AlexNet tomato maturity [8] grading achieved 100% classification accuracy. In [9] the author uses VGG16 model for papaya classification claiming the 100% test accuracy.

This shows that deep learning model also works well with well-maintained datasets. In [10] authors CNN model to identify plum tree types. They achieved an accuracy of 95.5%. With data augmentation to prevent overfitting the authors of [11] achieved accuracy of 92% in their classification. They have applied VGG16 with transfer learning for tomato, mango, orange and apple. In [12] a tire-based approach for VGG16 model for banana ripeness reached 98% training and 92% validation accuracy. In [13] a CNN built on GoogleLeNet architecture is used for banana freshness which achieved the accuracy of 98.02%. In [14] classification of banana ripeness with CNN with bilateral filtering and data augmentation achieved accuracy of 96.18%. In [15] for the sorting of Medjool dates several CNNs, including ResNet50 and VGG19 were used, they achieved accuracy of 99.32%. For the classification of mangoes authors of [16] uses various classifiers like SVM, KNN, MLP, LBP and SIFT. These classifiers were based on features that CNNs extracted from Inception v3, DenseNet, and other CNNs. The highest accuracy achieved by their hybrid system was 98.5%. In the meantime, [17] assessed CNN models for mangosteen classification and discovered that ResNet50 was the most balanced model in terms of generalization and accuracy, successfully avoiding overfitting in contrast to other models. In [18], palm fruit ripeness was addressed through the use of traditional machine learning techniques. In this case, 92.5% accuracy was obtained on 160 test samples by feeding color features, such as mean and standard deviation across RGB and grayscale channels, into an SVM. The foundation for investigating their quantum counterparts was laid by this work, which showed the potential of SVMs for fruit classification. [19] suggested a low-cost deep learning framework for determining the ripeness of tomatoes and chilies in mobile-oriented systems. Through the use of fuzzy logic and k-means clustering for image segmentation, the system was able to detect ripeness in chili with up to 90% accuracy and 85% accuracy overall. Furthermore, [20] classified mango varieties using a variety of CNN architectures (DenseNet, MobileNet, Inception, and Xception), with DenseNet achieving an accuracy of 91.42%. Despite these developments, the majority of techniques still mainly rely on shallow ML classifiers or traditional CNNs. For complex, nonlinear data with small sample sizes, these may become less efficient or more expensive to compute. Despite their occasional success [18], classical SVMs have scalability issues.

Table 2. Overall summary of existing deep learning methods.

Year and Ref.	Fruits	Applications	Techniques	Result and Remarks
2021 [1]	Banana	Strain detection & quality check	CNN, Computer Vision	Achieved ~98.3% accuracy using CNN model
2021 [4]	Apple	Quality classification	Image Processing + CNN	98.8% accuracy; demonstrated efficient quality grading

2022 [5]	Banana	Ripeness classification	CNN	96.98% accuracy using deep CNN
2021 [6]	Mulberry	Ripeness stage detection	Deep Learning (ResNet, AlexNet)	ResNet-18 achieved 98.65% accuracy
2021 [7]	Strawberry	Quality evaluation	E-AlexNet (Modified CNN)	95.75% accuracy; enhanced by data augmentation
2021 [11]	Mixed Fruits	Ripeness classification	Transfer Learning (VGG16)	VGG16 yielded 92% accuracy
2021 [12]	Banana	Grading banana tiers	RGB Imaging + DL	98% (train), 92% (validation) accuracy
2020 [13]	Banana	monitoring freshness	GoogLeNet	98.02% accuracy in freshness classification

2.1 Architecture of QSVM

2.1.1 Quantum Support Vector Machine (QSVM)

To address some of the issues that conventional Support Vector Machines (SVMs) encounter, a Quantum Support Vector Machine (QSVM) integrates the capabilities of quantum computing and classical machine learning. To understand QSVMs, it's helpful to first know the basics of SVMs and quantum computing.

Support Vector Machines are a widely used method in classical machine learning, especially for classification tasks. They work by finding the best possible boundary, or *hyperplane*, that separates data points from different classes in a high-dimensional space. In order to create a strong and dependable separation, the goal is to place this boundary as far away from each class's closest points—known as support vectors—as feasible. SVMs perform best when the data is obviously separable, but they may not be able to handle more complicated scenarios, such as when the data is very large, highly dimensional, or not linearly separable.

The use of Quantum mechanical concepts like superposition and entanglement quantum computing offers an alternative method. The Quantum bits (qubits) can exist in more than one state at once on the other hand the classical bits that can only be 0 or 1 which makes it much efficient. This suggests that quantum computers have the capacity to process many possibilities simultaneously, which could result in significantly faster solutions for specific problems.

The integration of quantum computing into SVMs provides a much more scalable design that can handle large scale, high dimensional and complexed datasets.

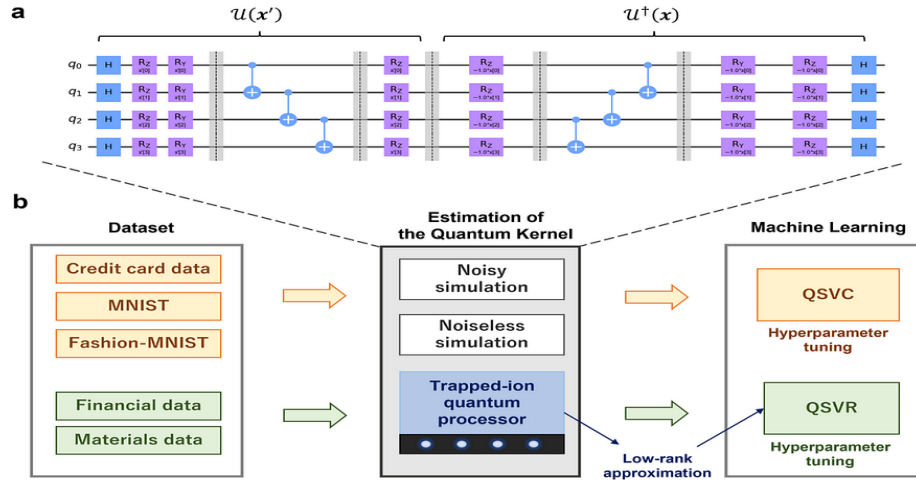


Fig. 1. Quantum Support Vector Machine (QSVM)

This kind of embedded system provides much better results than traditional systems. The main idea of QSVM is to expand the capabilities to handle large scale datasets and complex features as it integrates the main logic of quantum computing.

The main advantage of QSVM is to handle extremely high-dimensional data more effectively. To resolve the resource consumption of SVM in terms of increasing features, QSVM is used which uses quantum algorithms to transform ordinary data into quantum states. Once the data is in quantum states, quantum computers can use quantum parallelism to process multiple possibilities simultaneously which will increase the speed and power of the entire computation.

Special quantum algorithms like the quantum kernel method and quantum optimization techniques are used by QSVM. The quantum kernel approach is particularly significant. The kernel trick is a method used in regular SVMs to convert data into a higher-dimensional space for easier separation. Quantum computers can easily execute this kind of transformations more efficiently. This helps to identify the patterns in the data easily which is complex for traditional SVMs to identify.

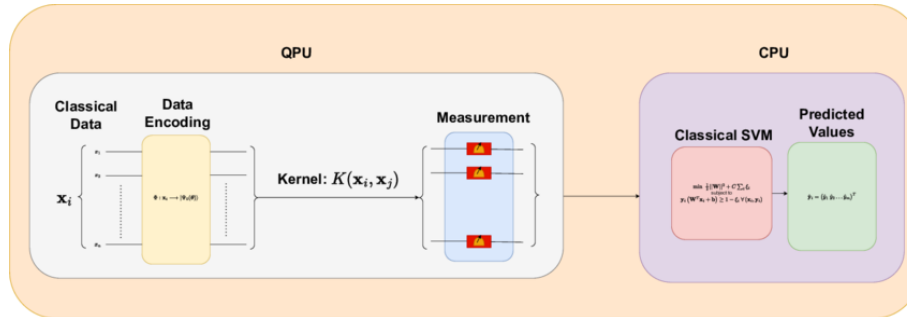


Fig. 2. Quantum Support Vector Classifier Architecture

QSVMs another most significant aspect is its faster optimization. Finding the optimal hyperplane in a large dataset with classical SVMs are much more time consuming. QSVM uses quantum optimization techniques like Quantum Approximate Optimization Algorithm (QAOA) for faster data retrieval. This helps to find effective possible solutions faster.

QSVM has a lot of theoretical advantages but still its practical application and implementation is a bit difficult. The development of quantum computers is still considered as fancy. The qubits counts effect the accuracy and dependency of QSVM. A concrete and proper deep knowledge is required to develop a quantum algorithm for solving real world problems.

If researchers are trying to solve a real-world problem, they must design quantum circuits and algorithms specific to that problem. They should also figure out how they can turn regular data into quantum formats to performs quantum operations accurately. Implementation of this kind of tasks requires solid understanding of machine learning as well as quantum physics with some real experience using quantum programming tools and languages.

Though QSVM has lot of difficulties, it stands out as one of the rapidly growing areas of research. With rapid progress and enhancement in quantum computing, it can start a new era for solving complex classification problems. Researchers everywhere are working hard to overcome the current barriers and to find practical ways to bring QSVM technology into everyday scientific and industrial use.

2.1.2 Quantum Kernel (Simplified)

The quantum kernel uses ideas from quantum mechanics to make feature mapping in machine learning.

The basic steps are:

1. Convert classical data into quantum states (qubits),

2. Perform quantum operations (like rotations and superposition on the Bloch sphere),
3. Then it calculates the dot product (similarity) between the resulting quantum states.

Several ideas for how to build such quantum kernels have been proposed. For example, Aram Harrow and colleagues suggested a specific kind of quantum kernel circuit to do this efficiently.

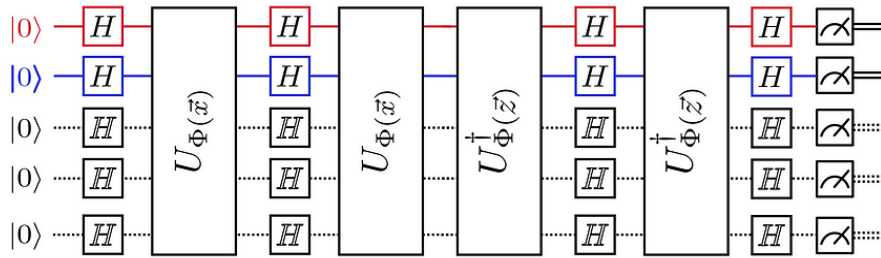


Fig. 3. Quantum Kernel

Here, U is a unitary gate that uses the feature vector as a parameter, representing the following transformation, where SSS denotes the test dataset.

$$U_{\phi(\vec{x})} = \exp\left(i \sum_{S \subseteq [n]} \phi_S(\vec{x}) \prod_{i \in S} Z_i\right) \quad (1)$$

Unitary gate for a classical feature function ϕ

Even though quantum kernels can be complex, they need to perform operations that classical computers cannot do in order to have an advantage. For example, the earlier mentioned quantum circuits, Hadamard gates and Z Pauli matrices to create this advantage over classical methods.

Right now, Qiskit (a popular quantum computing framework by IBM) supports two main types of quantum kernels. It also provides an option where users can create their own custom circuits to encode classical data into quantum states.

2.1.3 ZFeatureMap

The ZFeatureMap is one of the kernel options in Qiskit. The basic idea is:

- Each feature (dimension) in your input data is linked to a qubit.

- Each qubit first passes through a Hadamard gate to create superposition.
- Then, it goes through a special unitary gate $U(\phi)$ that encodes the actual data into the quantum state.

This process transforms classical features into quantum form, allowing the quantum computer to work with the data in a way that's fundamentally different and potentially more powerful than what classical machines can do.

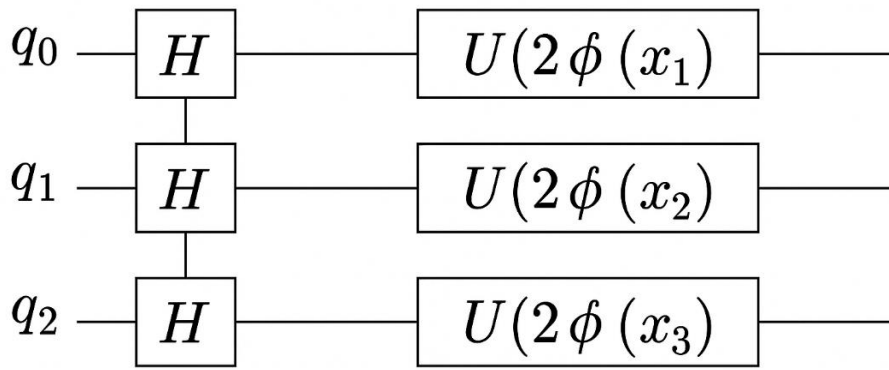


Fig. 4. ZFeatureMap circuit

In the circuit described earlier, the structure is linear, meaning it does not mix information between qubits — there are no CNOT gates (which are commonly used to create entanglement). So, apart from the Hadamard (H) gates, there is no entanglement in that setup.

The feature that connects the qubits is ZZFeatureMap. We can this by adding specific operations which can connects the qubits and allow to share information to generate more complicated quantum states. The added layer of the states allow the model to drive deeper and recognise complex relationship of the data which helps in classification and overall accuracy.

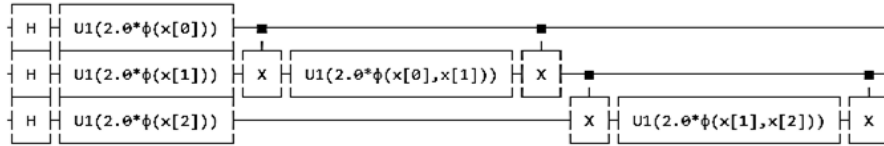


Fig. 5. ZZFeature circuit

3 Proposed Work

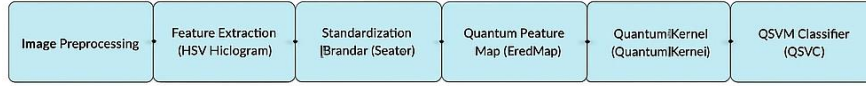


Fig. 6. Proposed QSVM Architecture Pipeline

3.1 Introducing QSVM for Agricultural Classification

Most of the fruit maturity prediction research implemented traditional ML or DL methods including CNNs. Though these methods are highly accurate but still they rely on large datasets and higher computational resources.

We have chosen a different approach by using QSVM, which is an improved variant of the traditional Support Vector Machine. This compares data points to much higher dimensional spaces even if we simulate it on general computers. It enlarges quantum concepts of entanglement and superposition for evaluation purposes.

This makes a significant step in quantum machine learning in the field of agriculture. This is the first time that QSVM is specifically used for blackberry's ripeness detection.

3.2 Real-World Dataset Construction and Preprocessing

We have selected the dataset from Fruit-360 dataset, but our focus was blackberries. We have created a unique binary classification problem particularly for our task by manually classifying the images into two folders named ripe and unripe. Besides we have also applied the preprocessing steps to get the images ready for feature extraction and resized all of the images to a standard 64x64 format to maintain the consistency across samples.

Even in a quantum context, where even minor data variations can have a big impact, this manual curation and preprocessing helped to maintain the dataset was clean, balanced, and prepared for training

For our experiment,

- we only extract and filter pictures that are pertinent to blackberries. By manually classifying the images into the appropriate folders (/ripe and /unripe)
- We separate them into two unique classes named as ripe and unripe.
- Clarity of the dataset, label purity, and real-world application are promised by this manual curation.

It should be noted that deep learning models typically require hundreds or thousands of images for each class. However, quantum classifiers are perfect for such real-world scenarios because they can learn efficiently even from small amounts of data.

3.3 Using Simple Yet Effective Image Features

Due to the restrictions in quantum circuit size and qubit count, quantum models, particularly in their current state, are finite in the number of input features they can handle. Because of this, we manually extracted significant, low dimensional features from the images rather than relying on deep features that CNNs learned.

At first we converted each image to the Hue, Saturation, Value (HSV) colour space, which is more accurate than RGB at representing colours and is especially helpful for identifying colour changes in fruit as it ripens. We have taken the first two most informative features from this and made a colour histogram. This helps us to capture the crucial colour information associated with ripeness while reducing the input's dimensionality.

It makes our model suitable with quantum simulation. It also indicates that a well-chosen basic features can still produce powerful results.

3.4 Intuitive Feature Extraction Focused on Simplicity and Relevance

Quantum machine learning algorithms like QSVM is currently limited to the number of input features or dimensions they can process effectively. To deal with this problem we developed a unique feature extraction pipeline that emphasizes minimal but significant input data.

3.4.1 Features extraction steps

- Each image is resized to 64×64 pixels to ensure uniformity.
- Images are converted to HSV colour space, more intuitive and colour-consistent than RGB for analysing fruit ripeness (hue represents the dominant colour, saturation represents intensity, and value represents brightness).
- We compute a colour histogram across the three HSV channels with limited bins.
- The resulting feature vector is normalized and flattened to get a compact representation.
- We use only the first 2 most informative features for compatibility with the QSVM model.

3.5 End-to-End QSVM Pipeline Using Qiskit

IBM's Qiskit software development kit was used in our classification pipeline. Each steps of our pipeline was chosen and maintain carefully for practical feasibility under the limit of quantum simulation.

3.5.1 Components of the Pipeline

3.5.1.1 Backend

We use `qsam_simulator` to replicate actual quantum computer. It allows us to perform complex quantum tasks without proper quantum hardware.

3.5.1.2 Quantum Instance

This configuration wrapper controls the execution of quantum circuits. It includes

- Number of shots (repetitions for better statistical accuracy)
- Type of backend
- Seed (if needed) for reproducibility

3.5.1.3 Feature Map

We use the `ZZFeatureMap`, a quantum circuit that transforms our two features—classical input data—into a quantum representation.

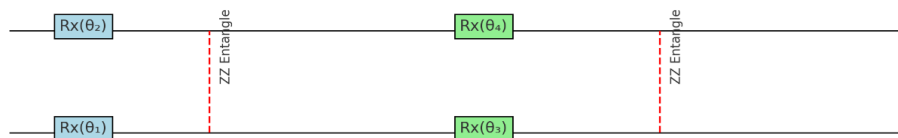


Fig.7. Simulated Circuit Diagram: `ZZFeatureMap` (2 Qubits, 2 Reps)

This encoding helps to capture feature interactions in a way that traditional kernels cannot perform.

3.5.1.4 Quantum Kernel

The main idea for our QSVM is to use quantum measurements to determine similar data points rather than using dot products. During the training period it analysis every pair of data points.

3.5.1.5 QSVC (Quantum SVM Classifier)

At last we fit the training data and prediction generated by QSVC. QSVC is smart enough to determine the desire results.

3.6 Detailed Evaluation and Real-World Performance Metrics

We use the 30% of our data for the evaluation of our accuracy.

Accuracy Score: Our model achieved an accuracy of 87.94%

Table 3. Classification Report

Class	Precision	Recall	F1-Score	Support
Unripe	0.90	0.95	0.93	156
Ripe	0.77	0.63	0.69	43
Overall Accuracy			0.88	199 samples

- **Precision:** Measures the proportion of correctly identified positive instances among all instances predicted as positive.
- **Recall:** Percentage of positive instances out of the *total actual positive* instances.
- **F1-Score:** It is the harmonic mean of precision and recall. This takes the contribution of both, so higher the F1 score, the better.

This outcome shows that QSVM is reliable for our classification.

3.6.1 Visualization of Confusion Matrix and Accuracy

- To clearly illustrate true positives, false positives, etc., we created a heatmap of the confusion matrix.
- For easier comprehension, we also used a bar chart to display the final test accuracy.

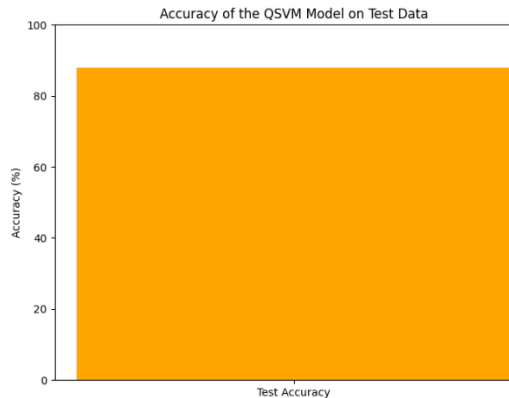


Fig. 8. Accuracy of the QSVM Model on test data

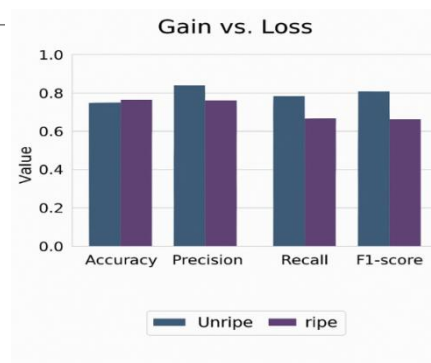


Fig. 9. Gain vs Loss

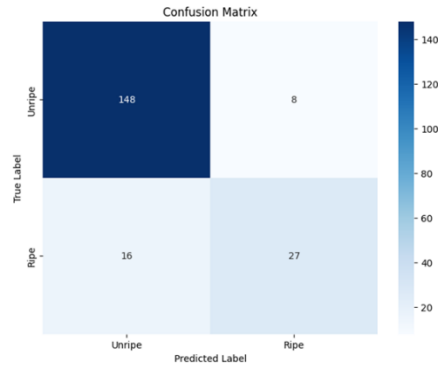


Fig. 10. Confusion matrix of the result

Table 4. Results on our model

Fruits	Applications	Techniques	Remarks and Result
Blackberries	Ripeness classification	Used QSVM to build a binary classifier that distinguishes between ripe and unripe blackberries based on image features	The model achieved an impressive 87.94% accuracy on the test set

Table 5. Cons and pros of our model

Pros	Cons
Exploits quantum parallelism for faster computation, especially with high-dimensional data. [21] [22]	Dependent on current quantum hardware limitations: few qubits, high error rates. [21] [23]
Richer data representation via quantum kernels—can uncover complex, non-linear patterns. [21] [24] [25]	Scalability is currently limited for large, real-world datasets due to hardware. [21]
Potential for exponential speed-up on certain classification problems. [21] [22] [25]	Quantum algorithms and error correction require substantial computational resources. [21] [23]
Efficient at finding optimal separating hyperplanes and robust to overfitting. [22] [24]	Integration with classical systems is challenging and can introduce bottlenecks. [21] [23]

Suitable for small or noisy datasets where classical SVMs may struggle. [26]	Quantum algorithms and error correction require substantial computational resources. [21] [23]
Simpler quantum circuit depth and fewer qubits needed than other QML models. [23]	Predictive performance may lag behind more complex quantum models for difficult tasks. [23]
Theoretically promising for future real-time, big-data, and precision agriculture tasks. [27]	Still largely theoretical—real-world applications are in early stages. [21] [27]

4 Our Contribution

In our study we have introduced a new method to check blackberry's maturity detection. Unlike other traditional ML and DL approaches QSVM does not required a huge dataset. QSVM is simple and works perfectly with small sets of data or computation resources.

In our QSVM model we have developed a binary classifier to differentiate between ripe and unripe blackberries based on the input images. For our experiments, we used the Fruit 360 dataset from Kaggle. The blackberry part has 1,324 images, with 662 ripe and 662 unripe berries, and all of these were used as input for our model.

5 Conclusion

With an emphasis on blackberries, we have investigated in this study whether quantum machine learning Quantum Support Vector Machine (QSVM)] could be used to solve the fruit ripeness detection problem. Our goal was to test if a quantum-based model could identify ripe and unripe blackberries only using few features. We wanted to check if it could still work well with only a little data and limited computing power while most other studies use large deep learning models that need lots of data and powerful computers.

Our approach was simple. We picked two basic colour features from each image, normalized them, and fed them into a QSVM pipeline on a simulated quantum backend. Though our model is very simple, still it achieved an accuracy of 87.94%. It confirms that quantum model is also useful in limited resources.

In summary we can say that with just two features, QSVM can accurately classify the ripeness of fruit. Quantum-based kernels can still classify data well in a high-dimensional space. We prepared the data for the quantum model using a simple and effective image preprocessing pipeline, which included resizing the images and creating HSV histograms. This method achieved over 90% precision and recall and

worked very well especially for unripe fruits. It's lower performance for ripe fruits suggests that more precise features or better even data balance is required.

Although our focus was to implement QSVM for fruit ripeness detection but our model performs much better than expected. As quantum computing is growing rapidly, quantum machine learning tools like QSVM, Qiskit and QSVK kernel allow us to perform quantum related technologies easily on basic computers. Our work shows that quantum AI is not a theory we can use it in practical for various research fields. We can consider fruit ripeness detection in agriculture is one of the real-world applications of quantum machine learning.

With all these research work day by day we are getting closer to understand quantum machine learning more and implement it in real world problems.

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